

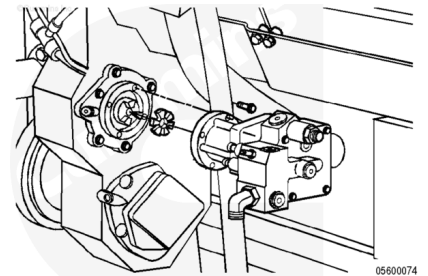
Trainings ▾

Install

with Mechanically Actuated Injector

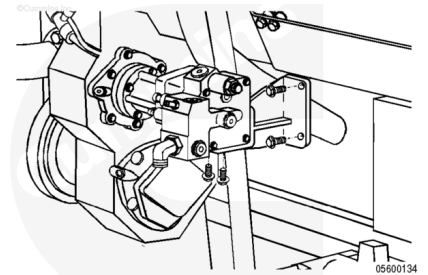
Install the fuel pump drive coupling, gasket, fuel pump, and four fuel pump mounting capscrews.

Completely install the fuel pump mounting capscrews, but do **not** tighten them.



Install the fuel pump support bracket to the bottom of the fuel pump and the engine block.

Tighten the support bracket mounting capscrews enough that the support bracket is in alignment.



Tighten the fuel pump mounting capscrews.

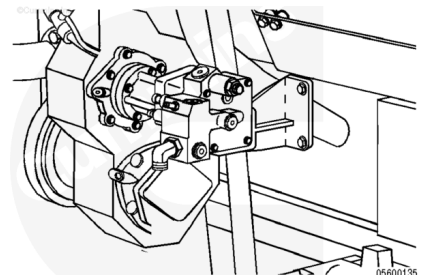
Torque Value: 45 n•m [33 ft-lb]

Tighten the fuel pump mounting bracket capscrews to the fuel pump.

Torque Value: 80 n•m [59 ft-lb]

Tighten the fuel pump bracket mounting capscrews to the engine block.

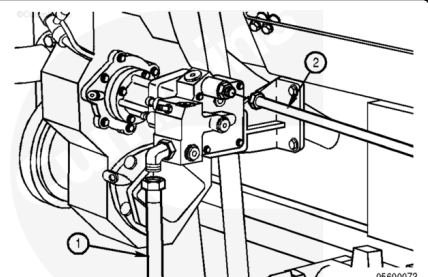
Torque Value: 80 n•m [59 ft-lb]



Install new o-rings in the fuel pump inlet hose fitting and the fuel supply hose fitting.

Install the fuel pump inlet hose (1) and the fuel supply hose (2).

Tighten the hoses.



Torque Value:

Fuel pump inlet hose 120 n•m [89 ft-lb]

Torque Value:

Fuel supply hose 120 n•m [89 ft-lb]

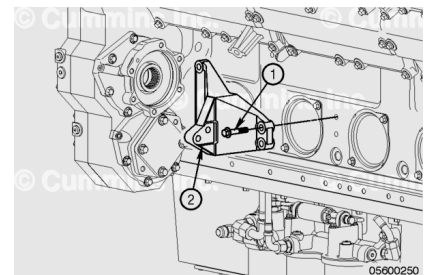
Connect the fuel pump actuator harness connection.

Connect the pressure sensor harness connection.

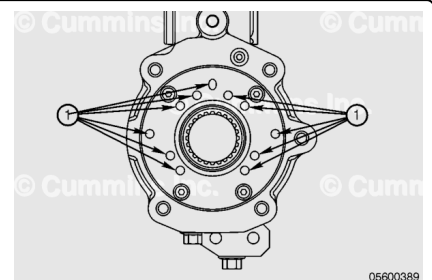
with Electronically Actuated Injector

Install the four capscrews (1) and fuel support bracket (2) to the engine block.

Hand-tighten the capscrews. Do **not** tighten at this time.

**⚠ WARNING ⚠**

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



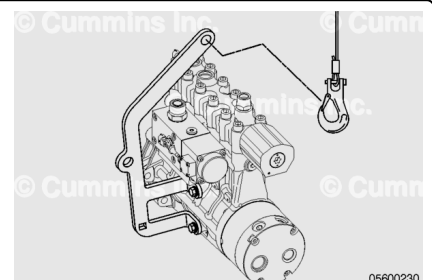
On a flat level surface, place the pump on its end with the gerotor-side down.

Pour 104 ml [3.5 oz] of fresh, clean lubricating oil into one of the eleven available face drillings (1).

Attach the lifting bracket to the fuel pump using the supplied hardware.

Torque Value: 45 n•m [33 ft-lb]

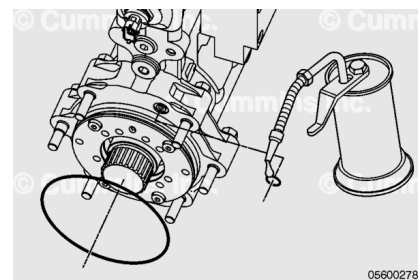
Attach the lifting hook to the lifting bracket.



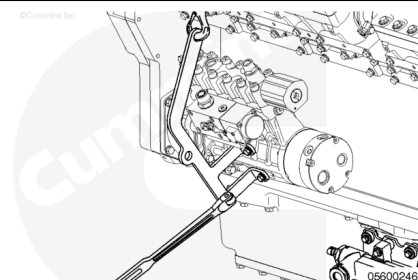
Use clean engine oil to lubricate both new o-rings.

Install the smaller o-ring that is used to seal the oil drilling from the fuel pump drive to the fuel pump.

Install the large o-ring that is used to seal the fuel pump drive to the fuel pump.

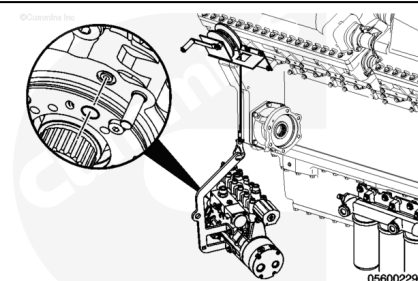


Attach a ½ inch breaker bar or ratchet to the notch in the lifting bracket.



⚠ CAUTION ⚠

Verify that the o-ring on the flange face of the fuel pump is in place. If the o-ring is missing, replace. Failure to do so will cause failure of the fuel pump due to lack of lubrication.

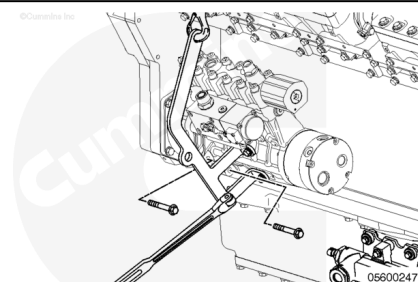


Raise the fuel pump into place using the hand winch.

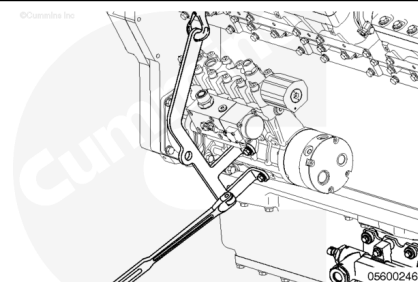
Install the fuel pump into the fuel pump drive. Use a breaker bar or ratchet to help guide the fuel pump into place.

Rotate the pump to align the mounting holes and install the six capscrews.

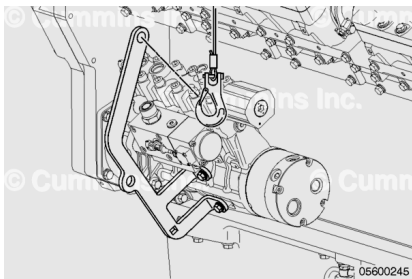
Torque Value: 117 n•m [86 ft-lb]



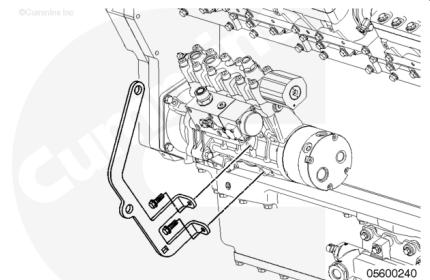
Remove the ½ inch breaker bar or ratchet from the lifting bracket.



Remove the hook from the lifting bracket.

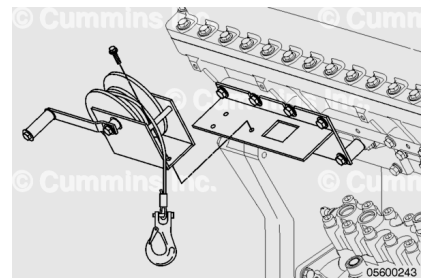


Remove the capscrews and lifting bracket from the fuel pump.

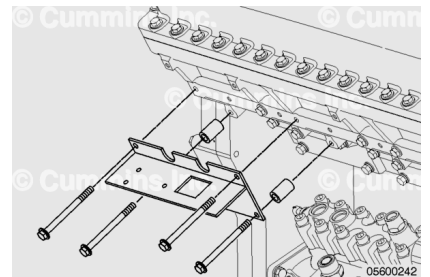


Remove the three capscrews, lock washers, and nuts attaching the winch to the winch support bracket.

Remove the winch.

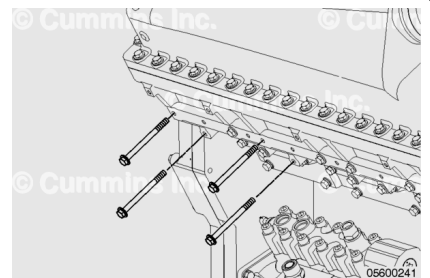


Remove the capscrews and winch support bracket.



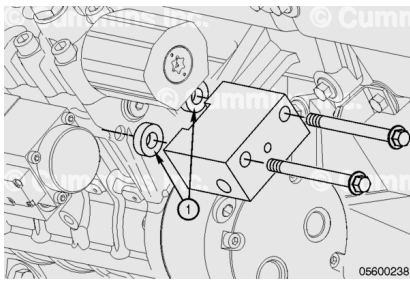
Install the intake manifold capscrews.

Torque Value: 68 n•m [50 ft-lb]



Install the two capscrews and spacers securing the upper fuel support bracket to the fuel pump.

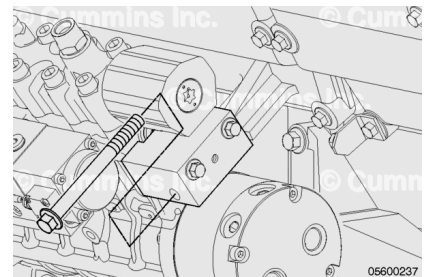
Hand-tighten the capscrews.



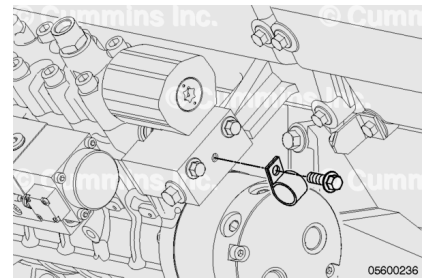
Install the cap screw securing the upper fuel support bracket to the camshaft cover.

Tighten the cap screws.

Torque Value: 180 n·m [133 ft-lb]



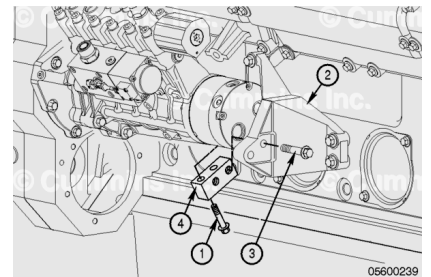
Install the P-clip securing the wiring of the metering valve to the upper fuel pump support.



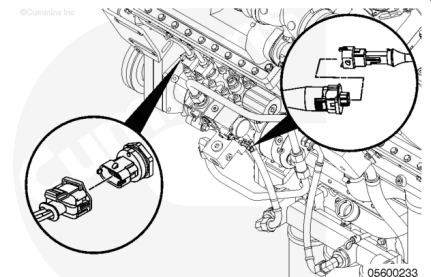
Install the two cap screws (3) securing the fuel pump support bracket (4) to the support bracket (2) mounted to the engine block.

Tighten the cap screws.

Torque Value: 113 n·m [83 ft-lb]



Connect the wiring harness to the fuel pump pressurizing assembly and injector metering rail 1 pressure sensor.



Install the intake manifold pressure sensors and intake manifold air temperature sensor that are located above the fuel pump. Use the following procedures in the QSK38, QSK50

and QSK60 (CM850 Modular Common Rail System)
Troubleshooting and Repair Manual, Bulletin 4021533.

- Refer to Procedure 019-061 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-061.html)
- Refer to Procedure 019-059 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-059.html)

